Single-Particle Tracking Reveals the Sequential Entry Process of the Bunyavirus Severe Fever with Thrombocytopenia Syndrome Virus

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The *Bunyavirales* is one of the largest groups of RNA viruses, which encompasses many strains that are highly pathogenic to animals and humans. Severe fever with thrombocytopenia syndrome virus (SFTSV) is an emerging tick-borne bunyavirus that causes severe disease in humans, with a high fatality rate of up to 30%. To date, the entry process of bunyavirus infection remains obscure. Here, using quantum dot (QD)-based single-particle tracking and multicolor imaging, the dynamic molecular process of SFTSV entry and penetration is systematically dissected. The results show that internalization of SFTSV into host cells is initiated by recruiting clathrin onto the cell membrane for the formation of clathrin-coated pits and further pinching off from the plasma membrane to form discrete vesicles. These vesicular carriers further deliver virions to Rab5+ early endosomes, and then to Rab7+ late endosomes. The intracellular transport of virion-carrying endocytic vesicles is dependent first on actin filaments at the cell periphery, and then on microtubules toward the cell interior. The final fusion events occur at ≈15–60 min post-entry, and are triggered by the acidic environment at ≈pH5.6 within the late endosomes. These results reveal the multistep SFTSV entry process and the dynamic virus–host interactions involved.

1. Introduction

The *Bunyavirales* is one of the largest groups of RNA viruses, which encompasses more than 350 members distributed among nine families.[1] The majority of bunyaviruses are arthropod-borne viruses that infect a wide range of hosts, including vertebrates, invertebrates, and plants. Four of the nine families comprise viruses that cause serious diseases with high mortality rates in animals and humans, including the highly pathogenic Crimean-Congo hemorrhagic fever virus (CCHFV) in the family *Nairoviridae*; the Rift Valley fever virus (RVFV), Toscana virus, and severe fever with thrombocytopenia syndrome virus (SFTSV) in the family *Phenuiviridae* (genus *Phlebovirus*); La Crosse virus (LACV) and Oropouche virus (OROV) in the family *Peribunyaviridae*; and several hantaviruses in the family *Hantaviridae*.[2] With an increasing frequency of outbreaks over the last decade, bunyaviruses are regarded as potential emerging pathogens that pose a global threat to livestock and human public health.[3]

SFTSV, an emerging bunyavirus in the genus *Phlebovirus*, was first identified in China in 2009.[4] SFTSV infection causes severe fever with thrombocytopenia syndrome (SFTS), with a high case fatality rate of up to 30%.[5] Since the emergence of this virus, SFTSV-infected patients have been identified in at least 23 provinces in China,[6,7] as well as in other countries such as Japan and South Korea.[8,9] There are currently no Food and Drug Administration (FDA)-approved vaccines or specific antivirals available against SFTSV owing to the limited information about the mechanisms underlying infection and pathogenesis of SFTSV. The rising incidence of SFTS disease in East Asia necessitates further investigation of the mechanisms involved in SFTSV infection.[10]

Viral entry into host cells, which is the first step of infection, represents a crucial target for therapeutic intervention. However, the entry pathways of bunyaviruses remain poorly characterized. Several studies based on the application of siRNAs, dominant-negative mutants, and chemical inhibitors suggest that LACV, OROV, and CCHFV mainly subvert clathrin-mediated endocytosis to invade cells.[11–13] For phleboviruses, the entry of the prototype member Uukunniemi virus (UUKV) is mainly clathrin-independent, as shown by clathrin silencing.[14] However, studies of the entry pathway of RVFV are controversial. A report suggested that the non-spreading strain of the virus was

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dependent on clathrin for entry,[15] while two other independent studies suggested that the RVFV vaccine strain MP12 entered cells through both a caveolin-dependent mechanism and micropinocytosis.[16,17] For SFTSV, there is few study about the entry process, except for a previous study showed that SFTSV entry could be blocked by dynasore, an inhibitor of dynamin, which is essential for the scission of clathrin-coated vesicles (CCVs), suggesting that clathrin may play a role in SFTSV entry.[18] Upon internalization, bunyaviruses, such as UUKV, RVFV, and CCHFV are delivered through early endosomes (EEs) to late endosomes (LEs) for acid-activated penetration.[14,15,19] In addition, it has been proposed that the cytoskeleton is involved in hantavirus and CCHFV entry.[20,21] Although some critical entry steps have been investigated, a comprehensive picture of the bunyavirus entry process remains to be depicted. There is an urgent need for developing effective approaches for unraveling dynamic entry process of bunyaviruses.

Single-particle tracking (SPT) in living cells not only contributes to visualization of the entry process at the single-virus level, but also facilitates the elucidation of the dynamic virus–host cell interaction with real-time evidence.[22] In recent years, owing to the superior brightness and stability of quantum dots (QDs), QD-based SPT has been extensively used for revealing virus infection process in living cells, such as the influenza virus, human immunodeficiency virus (HIV), and dengue virus.[23–25] In addition, a fluorescent lipophilic dye dequenching assay, a common method to visualize the membrane fusion, has been used for dissecting the viral fusion of Ebola virus and influenza virus.[26] To date, real-time visualization of the bunyavirus invasion process through QD-based SPT has not been reported. Here, we established QDs/DiD-based SPT technology for multi-color fluorescence monitoring the dynamic entry, penetration and virus–host interactions of SFTSV. Combined with biochemical approaches, the sequential processes involved in the entry of SFTSV, involving viral binding and internalization, vesicle trafficking, cytoskeleton transportation, and acid-triggered penetration, is revealed.

2. Results

2.1. SFTSV is Internalized via the Recruitment of Clathrin to the Plasma Membrane for Clathrin-Coated Pit (CCP) Formation

In order to visualize the course of SFTSV entry at the single-virus level in real time, SFTSV virions were biotinylated (bio-SFTSV) and purified by ultrafiltration. Initially, we tried to label bio-SFTSV with QDs through the in vitro labeling method,[27] however it was not successful. Therefore, we exploited an alternative method.[28] Bio-SFTSV was bound to the Vero cells surface and incubated with streptavidin-modified QDs (SA-QDs) in situ, then extensive washing with phosphate-buffered saline (PBS) were used to remove unbound virions and QDs. Transmission electron microscopy (TEM) showed that the purified bio-SFTSV particles were structurally similar to native virions (**Figure 1**A). The specificity and efficiency of QDs labeling were examined by analyzing colocalization efficiency of QDs signals (red) with immunofluorescence signals (green) using anti-SFTSV-nucleoprotein (NP) antibody (Figure S1, Supporting Information). The result showed that biotinylated virions were efficiently labeled with SA-QDs (Figure S1, upper panel, Supporting Information). In contrast, fluorescence signals of QDs were rarely observed in the control group (SFTSV without biotinylation), suggesting that there was neither obvious nonspecific adsorption of SA-QDs onto cells, nor between SA-QDs and unbiotinylated SFTSV (Figure S1, lower panel, Supporting Information). The infectivity of bio-SFTSV and QDs-SFTSV was tested with the parental SFTSV as a control; the results showed that neither biotinylation nor conjugation with SA-QDs substantially impaired the infectivity of SFTSV (Figure 1B). Therefore, SFTSV particles were labeled with QDs efficiently and could be used in subsequent experiments.

The classical endocytic mechanisms mainly include clathrin-dependent endocytosis, caveolae- mediated endocytosis, and macropinocytosis. Although a previous study using inhibitors of dynamin suggested that the entry of SFTSV may be clathrin-dependent,[18] the detailed mechanisms involved in SFTSV endocytic uptake remained unclear. Here, we exploited dual-color fluorescence imaging to visualize the entry of QDs-SFTSV into Vero cells which transiently expressing an enhanced green fluorescent protein labeled clathrin light chain (EGFP-Cla). Interestingly, we observed that SFTSV infection led to recruitment of clathrin onto the cell surface (Figure 1C). Quantification of SFTSV-infected cells (n = 20) showed that 46.8% of clathrin puncta were detected on the cell membrane in the presence of SFTSV binding (Figure 1D). However, in the absence of virus infection (n = 20), clathrins exhibited a scattered distribution in the cytoplasm, with only 19.5% of green puncta located on the cell surface (Figure 1D). SFTSV infection also promoted the formation of clathrin-coated vesicles (CCVs). The number of CCVs (Figure 1C, right, white arrows) in SFTSV infected cells was ≈8 fold higher than in uninfected cells (Figure 1C, left), as Figure 1E shown.

Successive snapshots captured that virus particles (n > 10) initially bound to clathrin structures on the cell membrane, gradually pinched off from the plasma membrane, and entered into the cytoplasm together with part of the clathrin cluster (Figure 1F, upper panel and Movie S1, Supporting Information). The rest of the clathrin cluster that remains on the plasma membrane may continuously recruit clathrin to produce new CCVs at the same site.[25] The CCVs containing QDs-SFTSV (n = 20) were observed to move from the plasma membrane toward the interior of the cells (Figure 1F, lower panel and Movie S2, Supporting Information). Ultrastructural observation also revealed the clathrin-mediated entry of SFTSV. As shown in Figure 1G, at 0 min postinfection (p.i.), a SFTSV particle bound to the plasma membrane (1#). At 5–10 min p.i., virions were internalized by a shallow, dome-shaped invagination of the plasma membrane that lacked obvious cytoplasmic coats (2#), and then by a more pronounced pit with a visible CCP (3#, white arrow). At 15 min p.i., an internalized SFTSV was observed within a CCV (4#). To further confirm the role of clathrin-mediated endocytosis in the entry of native, non-biotinylated SFTSV, two well-known inhibitors, chlorpromazine (CPZ) and sucrose, which inhibit the assembly of CCPs/CCVs, were used. The results showed that both CPZ and sucrose inhibited SFTSV infectivity (Figure S2B, titration assay, Supporting Information) and viral protein expression (Figure S2B, Supporting Information, Western blotting analysis) in a dose-dependent manner (Figure S2B, Supporting Information). In addition, dynasore, a drug that inhibits dynamin activity, was

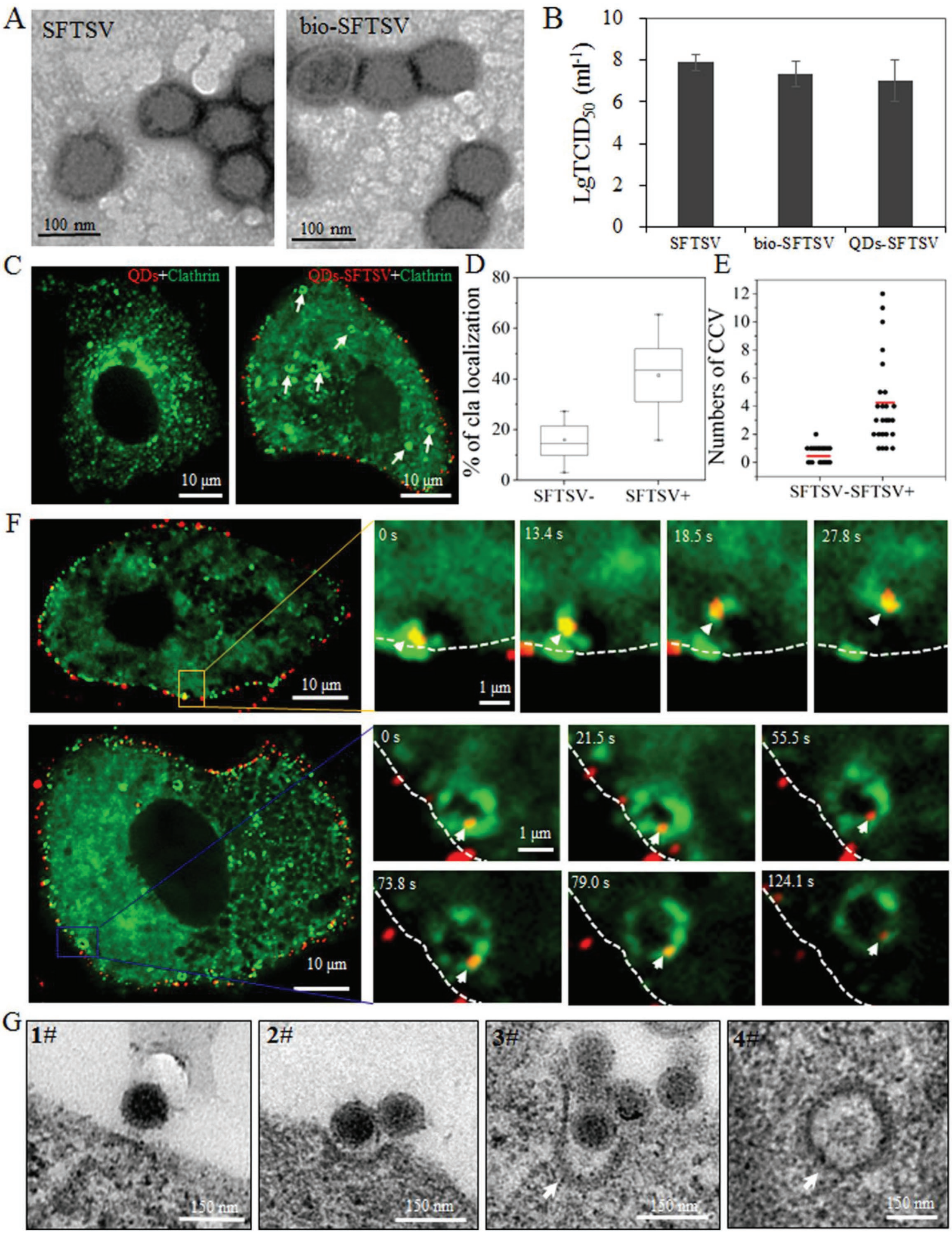


Figure 1. SFTSV entry is clathrin-dependent. A,B) Characterization of QDs-labeled SFTSV. A) The purified SFTSV and bio-SFTSV were negatively stained and examined by electron microscopy. B) Viral infectivity assay. Vero cells were infected with SFTSV, bio-SFTSV or QDs-SFTSV and collected at 48 h post infection (h p.i.) for virus titration assays. C) SFTSV (red) binding recruited clathrin (green) to the membrane and induced the formation of CCVs (indicated by white arrows). D) Percentage of clathrin located on the plasma membrane in the SFTSV-infected or uninfected cells as shown in (A), $n = 20$ for each group. E) Numbers of CCVs in the SFTSV-infected or uninfected cells. The numbers of CCVs in the uninfected and infected cells ($n = 22$ for each group) were measured and the average numbers (the red bars showed) of CCVs per cell in the two groups was calculated, respectively. F) SFTSV was internalized into host cell through reorganization of clathrin. Upper panel, series of images of a QDs-SFTSV particle internalized together with a clathrin cluster (arrow) (0–27.8s). Lower panel, snapshots of an internalized QDs-SFTSV particle delivered within a CCV (arrow) toward the cell interior (0–124.1s). The white dashed lines indicate the plasma membrane. G) Ultrastructural analysis of entry of SFTSV.

also found to block SFTSV infection (data not shown). These results suggest that clathrin is required by SFTSV infection, which results in the recruitment of clathrin to the plasma membrane for CCP formation, followed by viral invagination into host cells through the detachment of CCPs from the plasma membrane.

The potential roles of caveolae-mediated endocytosis and macropinocytosis in SFTSV entry were also investigated. Colocalization signals of QDs-SFTSV and caveolin were rarely found, suggesting a very low probability of their colocalization (Figure S2A and Movie S3, Supporting Information). When nystatin or phorbol 12-myristate 13-acetate (PMA) was used to block caveolae-mediated endocytosis, the infectivity and the viral protein expression of the progeny virus was not significantly reduced (Figure S2C, Supporting Information). Similarly, inhibitors of macropinocytosis by IPA3 (p21-activated kinase inhibitor III) and RAC-1 (RacGEF inhibitor NSC23766) also did not obviously impair SFTSV infection (Figure S2D, Supporting Information). The image data obtained from QDs-SFTSV was consistent with the results from biochemical assays and ultrastructural analysis by using native SFTSV (Figure 1C–G and Figure S2B–D, Supporting Information), further indicating neither biotinylation nor QD labeling changed the mechanisms of SFTSV entry. Taken together, these data demonstrate that SFTSV entry is dependent on clathrin-mediated endocytosis rather than on caveolae-mediated endocytosis or macropinocytosis.

2.2. SFTSV is Delivered from CCVs to Rab5-Positive EEs then to Rab7-Positive LEs

Following internalization into the cytoplasm via CCVs, viral particles are usually sequestered in a series of endocytic vesicles for further intracellular transportation before release of the virus genome. The proportions of SFTSV particles colocalized with Rab5 or Rab7 to the total particles in the cell was measured, and the results demonstrated that ≈30%, 45%, and 34% of SFTSV particles were colocalized with Rab5-positive EEs (Rab5+ EEs) at 30, 45, and 60 min p.i., respectively, while 17%, 33%, and 41% of virions were colocalized with Rab7-positive LEs (Rab7+ LEs) at 30, 45, and 60 min p.i., respectively ($n \geq 20$ for each group) (**Figure 2**A,B). The results indicated that the delivery of SFTSV in the cytoplasm involves sequential transfer from Rab5+ EEs to Rab7+ LEs. As endosomes mature, they grow increasingly acidic, which plays a vital role in triggering endosomal membrane penetration of viruses. Therefore, chloroquine (CQ) and NH_4Cl were used to inhibit the acidification of endosomes for determining the role of endosomes in viral infection. In cells treated with CQ or NH_4Cl, the infectivity of the progeny viruses was significantly decreased (Figure S2E, Supporting Information), suggesting that the luminal acidic pH of endosomes is required for SFTSV infection. Furthermore, TEM also confirmed the SFTSV was delivered from EEs to LEs. Normally, EEs are characterized as a complex structure, with both tubular and vacuolar domains, or as round, turgescent, and empty organelles with few intraluminal vesicles. In contrast, mature LEs, which are also known as multivesicular endosomes (MVEs), are typically 250–1000 nm in diameter and contain numerous intraluminal vesicles; a proportion of these LEs with partial electron dense areas were identified as endolysosomes (ELs).[29–31] At 10–15 min p.i., single particle was occasionally seen in EE-like structures with both tubular and vacuolar domains (Figure 2C, EE 1#); however, they were frequently detected in smooth-surfaced vacuolar vesicles (EE 2#). At 15–30 min p.i., SFTSV was observed in classical early endosomal vacuoles with diameters of 400–500 nm (EE 3#). At 30–60 min p.i., virions were often observed in MVE with a few internal vesicles, LE containing numerous internal vesicles or multilaminar membrane structures or in EL-like structures with partial electron dense areas (Figure 2C, lower panel).

To trace SFTSV endocytic trafficking processes, we performed real-time multicolor fluorescence-tracking of virus particles and endocytic vesicles. As shown in **Figure 3**A and Movie S4, Supporting Information, a SFTSV particle (blue) was encompassed by a vesicle containing both clathrin (green) and Rab5 (red); this was followed by the gradual shedding of the clathrin coat (white arrow) from the vesicle and simultaneous recruitment of Rab5 (white arrowheads) onto the vesicle membrane. To normalize the influence of the photobleaching, the ratio (Rr or Rg) of the red or green fluorescence intensity in the vesicles to the corresponding fluorescence intensity in the whole cell was analyzed. The increased Rr/Rg (R/G) ratio (Figure 3A) additionally indicated attenuation in the fluorescence intensity of clathrin, while stepwise increase in Rab5 fluorescence signals, suggesting the CCV was developing into an EE. Similar phenomena were observed in four vesicles with one shown in Figure 3A. Another set of snapshots captured a SFTSV particle (blue) located within a vesicle decorated with both Rab5 (red) and Rab7 (green) (Figure 3B and Movie S5, Supporting Information). Similarly, the decreasing R/G ratio of four vesicles (Figure 3B, only one representative endosome was shown) demonstrated gradual weakening and step-wise strengthening of the fluorescence signals of Rab5 and Rab7, respectively. These findings suggest that Rab5 was gradually lost and replaced with Rab7, which ultimately triggered the switch from Rab5+ EEs to Rab7+ LEs. These results demonstrate that CCVs deliver SFTSV particles and membranes to EEs and then to LEs.

2.3. SFTSV Transport is Dependent on Actin Filaments at the Cell Periphery and Microtubules toward the Cell Interior

Real-time visualization was used to monitor virus moving path and dynamics in cytoplasm (Figure S3 and Movie S6, Supporting Information). The initial motility of the virus was relatively slow and restricted during passing through the membrane, while upon entering into the cell, the virus particle experienced a relatively quick and directed movement at the host cell periphery and in the cell interior region, indicating the virus likely transported along the cytoskeleton. Furthermore, we investigated whether the cytoskeleton is involved in SFTSV trafficking. Dual-color fluorescence observation of SFTSV and actin filaments showed that before internalization, virus particles bound to actin-enriched filopodia-like protrusions and moved along them to reach the cell surface, at a relatively rapid speed (mean 0.17 ± 0.15 μm s^{-1}, $n = 10$),

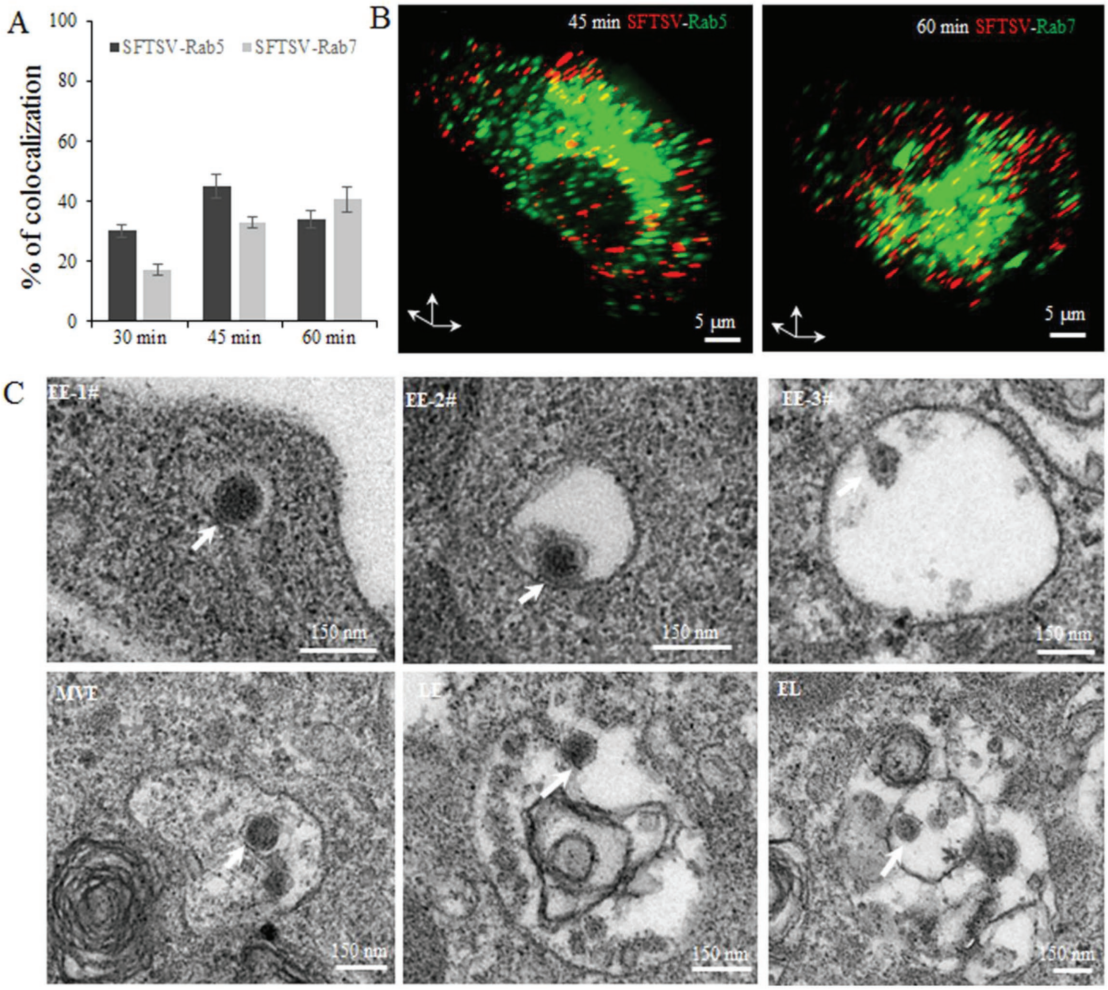


Figure 2. SFTSV enters early endosomes and late endosomes. A,B) Colocalization analysis of SFTSV with endosomes. A) The proportion of SFTSV particles colocalized with Rab5 or Rab7 to the total particles in the cell in the three-dimensional (3D) confocal images at 30, 45, and 60 min p.i. ($n \geq 20$ for each group). B) An example of SFTSV colocalized with Rab5 at 45 min p.i. and with Rab7 at 60 min p.i. 3D confocal images of SFTSV particles (red) and EEs or LEs (green) were shown. C) Ultrastructural micrographs of the internalized SFTSV particles located in endosomes.

and then slowed down near the root of the protrusions close to the cell surface (**Figure 4**A,E, and Movie S7, Supporting Information). TEM studies also showed viral particles adhered to the filopodia-like structure extending from the cell surface (Figure 4B). At the cell surface, SFTSV moved continuously along the actin filament at a relatively slow speed (mean 0.08 ± 0.07 μm s^{-1}, n = 10) (Figure 4C,F, and Movie S8, Supporting Information). Dual-color fluorescence observation of SFTSV and microtubules showed that, following internalization, SFTSV moved along microtubules toward the central cytoplasm (Figure 4D and Movie S9, Supporting Information). The movement of SFTSV along microtubules was initially relatively rapid (mean 0.32 ± 0.14 μm s^{-1}, n = 10) (Figure 4D,G, 0–25.7 s), and deceleration was observed near an intersection of microtubules (Figure 4D,G, 25.7–102.6 s). On fitting the relationship between mean square displacement (MSD)-time interval ($n\Delta t$), the movement was found to be directed motion (data not shown). The data demonstrated that the cytoskeleton-dependent intracellular transport of SFTSV initially depended on actin filament-dependent trafficking at the cell periphery and then on microtubule-dependent transport toward the cell interior region.

To confirm the roles of actin filaments and microtubules in SFTSV transport, the effects of cytochalasin D (CD, inhibits actin filament elongation at the barbed end), 2,3-butanedione monoxime (BDM, myosin motor inhibitors) and nocodazole (microtubule-depolymerizing agent) on SFTSV infection were analyzed. Compared with dimethyl sulfoxide (DMSO), both CD and nocodazole caused a concentration-dependent decrease in SFTSV infections (Figure S2F, Supporting Information). The BDM had a similar inhibitory effect on SFTSV infection (data not shown). Furthermore, the motility of SFTSV particles was investigated after treatment with nocodazole or CD (n > 10 for each group). In the untreated cells, the average instantaneous velocities of SFTSV were 0.16 ± 0.19 μm s^{-1}, with the highest speed reaching 1.6 μm s^{-1}. Some particles experienced directed (**Figure 5**A, white arrowhead) or bidirectional directed (Figure 5A, white arrows) long-distance movements to reach the cell interior, covering 14.88 ± 9.2 μm mean linear distance from the starting point to the end site within

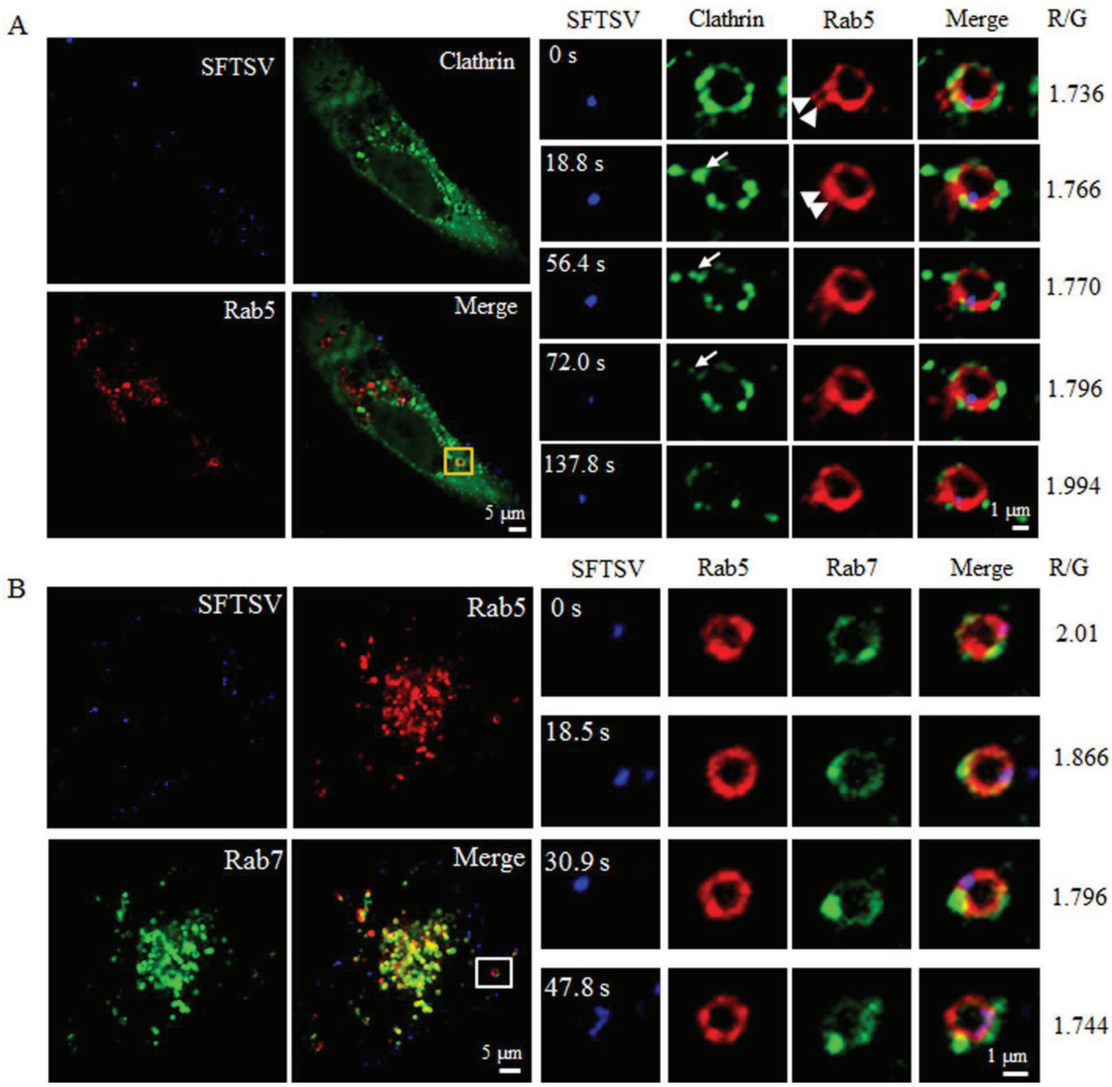


Figure 3. CCVs encompassing SFTSV maturated to EEs and then to LEs. A) CCVs deliver viruses and vacuolar domains to Rab5+ EEs. Multi-color images of QDs-SFTSV (blue), clathrin (green), and Rab5 (red) simultaneously were captured under a cell culture system at 15 min p.i.. A series of images is shown of CCV containing an SFTSV particle shedding its clathrin coat (white arrow), and simultaneously recruiting Rab5 (white arrowheads) onto the vesicle membrane to form the Rab5+ EEs. B) EEs containing SFTSV maturing to LEs; Snapshots of SFTSV particles (blue) within a Rab5+ (red) and Rab7+ (green) vesicle at 30 min p.i., showing a gradual weakening of the fluorescence intensity in Rab5, while showing stepwise strengthening in Rab7. *R/G* = Rr/Rg, where the Rr and Rg represents the ratio of red and green fluorescence intensity in the vesicles to corresponding fluorescence intensity in the whole cell, respectively.

5 min (n = 11) (Figure 5A,B,E, untreated, and Movie S10, Supporting Information). In contrast, after treatment with CD, most SFTSV particles showed incessant "trembling" in a limited area near the plasma membrane with relatively lower speed (0.07 ± 0.08 μm s^{-1}), and shorter distance (mean linear distance of 4.828 ± 2.2 μm) (n = 10) (Figure 5A,C,E, CD, and Movie S11, Supporting Information). In nocodazole-treated cells, the SFTSV particles underwent short-distance directed movements (mean linear distance of 7.28 ± 3.6 μm), with mean speeds of 0.09 ± 0.07 μm s^{-1} (n = 10) (Figure 5A,D,E, nocodazole, and Movie S12, Supporting Information). Most particles were trapped at the cell periphery without deeper penetration into the cell interior; this was putatively attributed to the failure of switchover from the actin filaments to microtubules. These results confirm that both actin filaments and microtubules are required for the cytoplasmic trafficking of SFTSV.

2.4. Endosomes Carrying SFTSV Move along Microtubules toward the Cell Interior

As internalized SFTSV virions were found to be delivered through endocytic vesicles, we further monitored the transport behaviors of virus-containing endosomes by fluorescence imaging. Analysis of ten virus-containing endosomes suggested that Rab5+ endosomes could mainly undergo long-distance directed motion (Figure S4A,B, Supporting Information); A typical trajectory of a virus-containing Rab5+ endosome

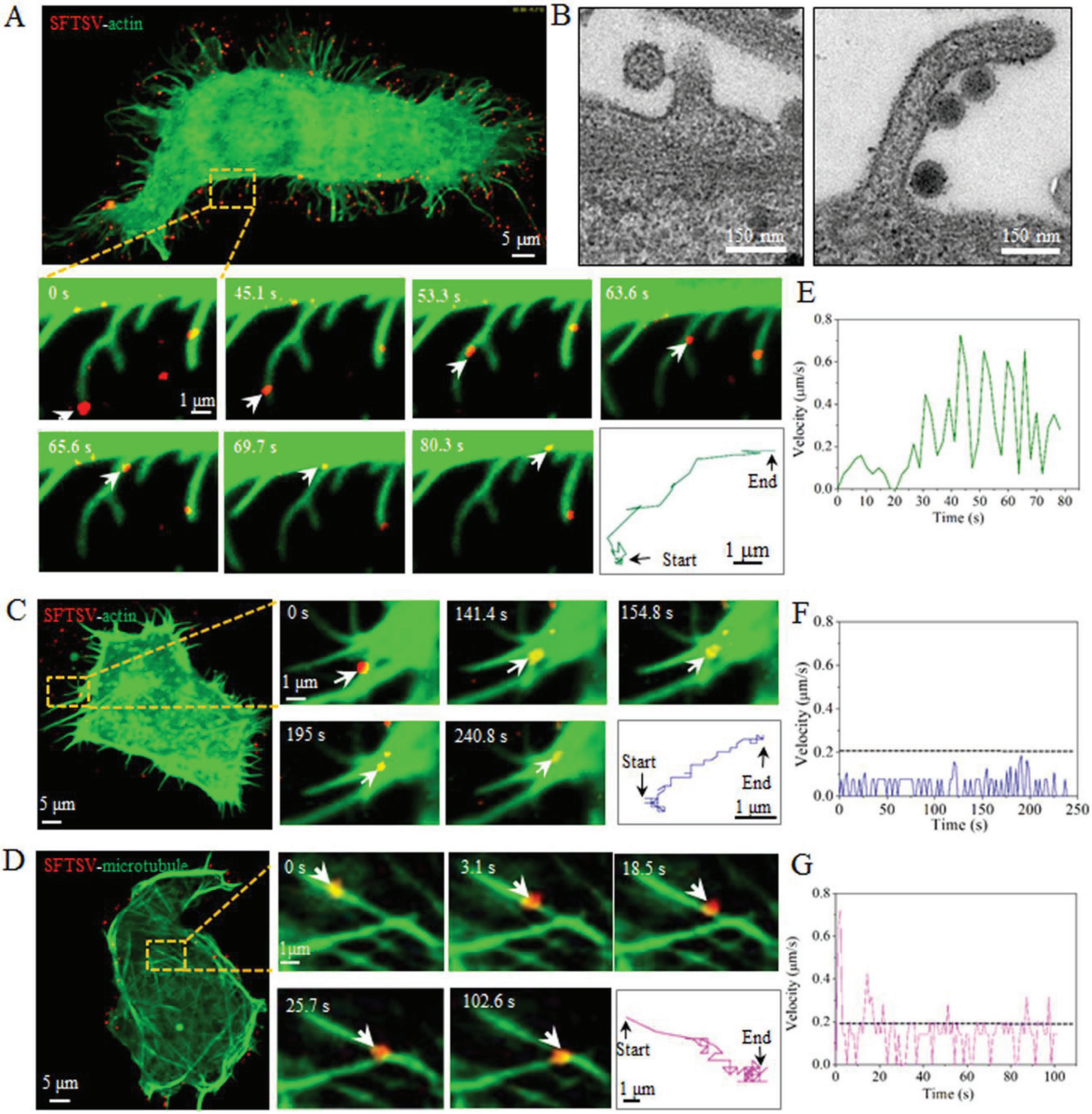


Figure 4. SFTSV moving along actin filaments and microtubules. A) SFTSV transport along actin-enriched filopodia-like protrusions. Snapshots of a SFTSV particle (red, white arrow) was transported along the actin-enriched protrusions (green) reaching the cell surface (0–80.3s). The green line indicates the trajectories of SFTSV movement. B) TEM image of SFTSV binding to the filopodia-like protrusions extending from the cell surface. C) SFTSV moving on the actin filament in the cell periphery. Snapshots showing a SFTSV particle (white arrow) being transported along the actin filament at the cell periphery (0–390s). The blue line indicates the trajectories of SFTSV movement. D) SFTSV moving along the microtubule in the cell interior. A series of images of a SFTSV particle (red, white arrow) being transported along a microtubule (green) in the cell interior. The pink line indicates the trajectories of SFTSV movement. E–G) Instantaneous velocities of the virus particles shown in (A,C,D), respectively.

moving from the cell periphery to the cell interior is shown in **Figure 6**A and Movie S13, Supporting Information; the trajectory and MSD-$n\Delta t$ plots suggested that their transport process may be dissected into two stages: initially, a restricted motion at the periphery region (Stage I), and then bidirectional movement directed toward the interior of the cell (Stage II). In contrast, the tracing of ten virus-containing Rab7+ endosomes showed that the majority of Rab7+ endosomes primarily underwent relatively slow and restricted motion in the cell interior region (Figure S4A,C, Supporting Information). As shown

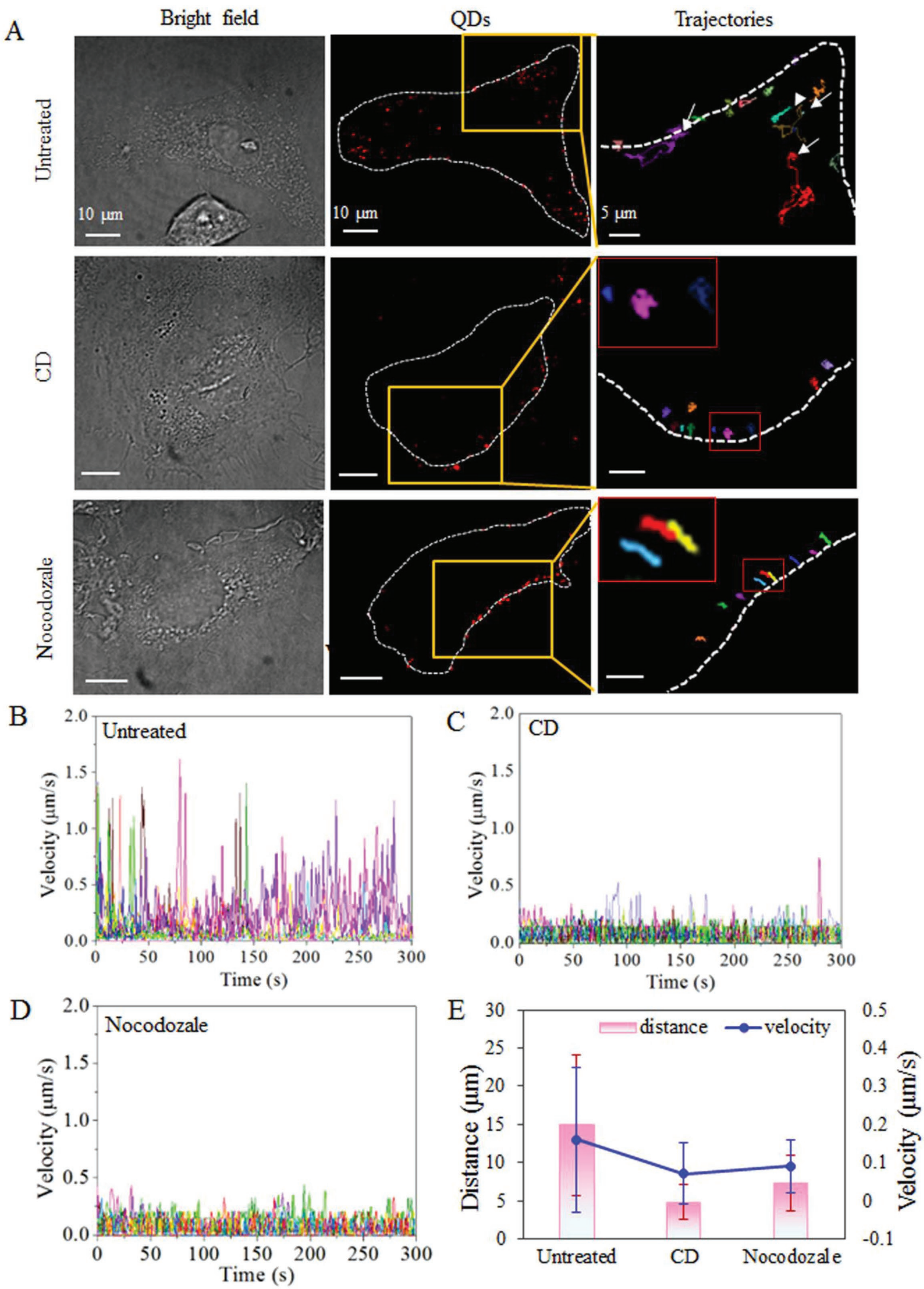
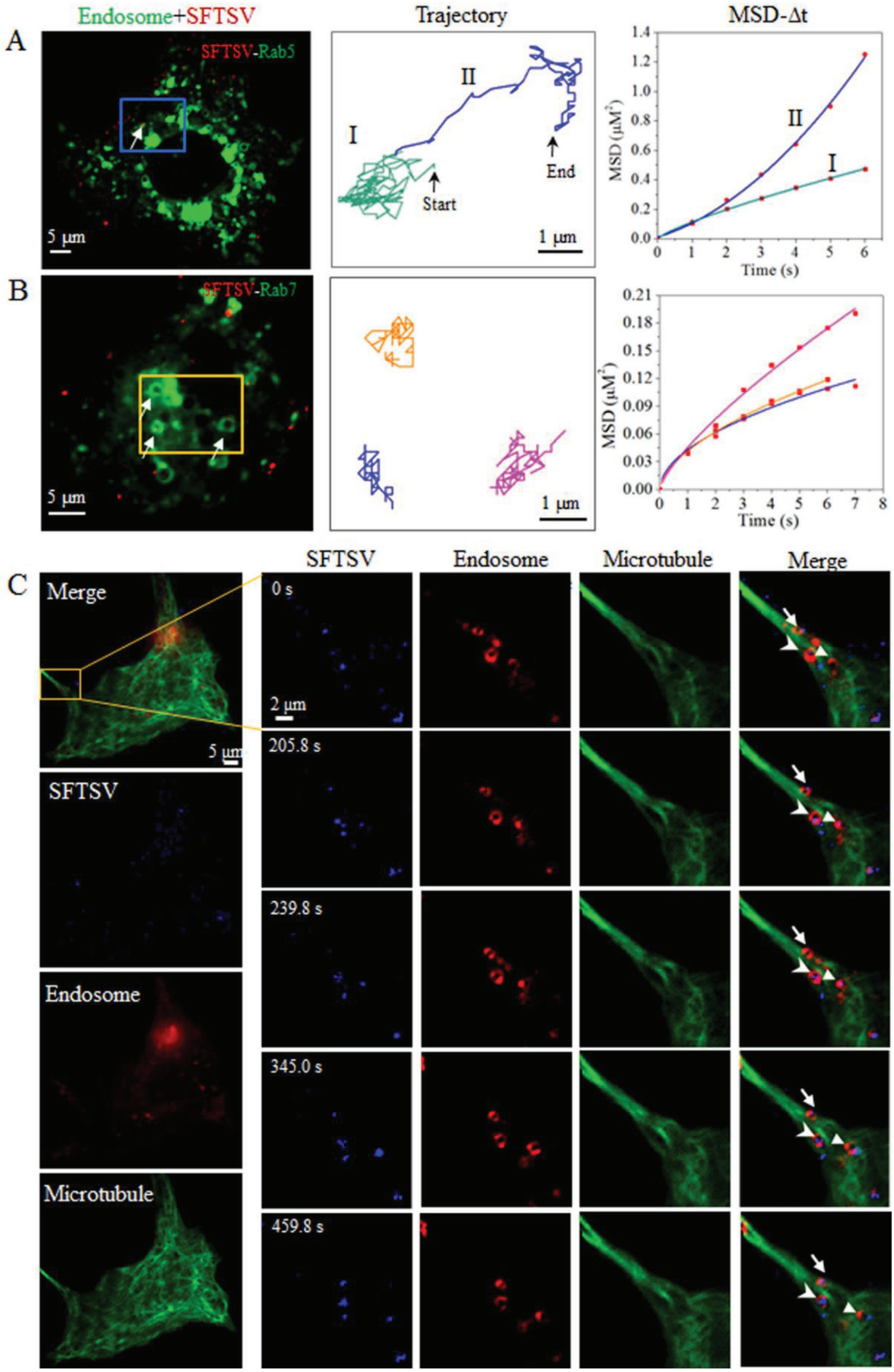


Figure 5. SFTSV transport is dependent on actin filaments and microtubules. A) Motion of SFTSV in untreated or CD- or nocodazole-treated host cells. The following are shown: a bright-field image of a cell, confocal images of QDs-SFTSV (red), the trajectories of SFTSV movement in the left boxed regions in the middle image (QDs). In untreated cells, particles experienced directed (white arrowhead) or bidirectional directed (white arrows) long-distance movements toward the cell interior. The white dashed lines indicated plasma membranes. B–D) The instantaneous velocities of SFTSV particles transport in the cells B) untreated or C) treated with CD or D) nocodazole shown in (A). E) The average distances and velocities of SFTSV particles movement in the cells B) untreated or C) treated with CD or D) nocodazole shown in (A). The error bars represent standard deviation (SD).


Endosome+SFTSV
Trajectory
MSD-Δt
A
SFTSV-Rab5
5 μm
I
II
Start
End
1 μm
MSD (μM²)
Time (s)
B
SFTSV-Rab7
5 μm
1 μm
MSD (μM²)
Time (s)
C
Merge
5 μm
SFTSV
Endosome
Microtubule
Merge
0 s
2 μm
205.8 s
239.8 s
345.0 s
459.8 s

in Figure 6B and Movie S14, Supporting Information, three representative trajectories of Rab7+ endosomes showed that the virus-containing Rab7+ endosomes underwent short-range back and forth restricted motion in the cell interior.

Next, we investigated whether SFTSV-carrying endosomes moved along the microtubules from the cell periphery toward the cell interior. Since the above results showed that Rab5+ endosomes underwent a more rapid, directed, and longer distance movement compared with Rab7+ endosomes (Figure 6A,B, and Figure S4A–C, Supporting Information), we visualized the dynamic interactions between SFTSV-carrying Rab5+ endosomes with microtubules by multicolor fluorescence tracking. The results clearly showed that virus-containing Rab5+ endosomes transported along microtubules from the cell periphery to the interior region (Figure 6C, white arrow and arrowhead, Movie S15 and Figure S4D–E, Supporting Information).

2.5. SFTSV Penetration is Triggered by Low pH of ≈5.6 within LEs

To visualize the fusion of SFTSV particles with endosomes, a fluorescent dye dequenching assay, a commonly used method to signal viral, vesicular, and cellular fusion events, was employed.[26] In this study, DiD was chosen as a fluorescent lipophilic dye that spontaneously incorporates into the viral membrane and exhibits fluorescence self-quenching at high concentrations, and a significant increase in fluorescence intensity when viruses fuse with endosomes. QDs-SFTSV was labeled with DiD (QDs-DiD-SFTSV), incubated with cells at 4 °C, and then shifted to 37 °C to allow synchronized infection. Upon viral binding, virus particles were mainly observed as red spots (QDs signals) at the cell surface from 0 (data not shown) to 10 min (**Figure 7**A, 10 min). After a 15 min incubation, viral fusion with endosomes was evident by fluorescence dequenching showing obvious yellow fluorescence signals (merged signals of QD and DiD) (Figure 7A). From 15 to 60 min, the fluorescence intensity of DiD increased, while the QDs signals decreased with time (Figure 7B, top and middle). Quantitative analyses showed that at 15 min p.i., the rate of fused viruses was about 20% ant then increased to 35% and 64% at 30 and 60 min p.i., respectively (n > 15 cells for each group) (Figure 7B, lower panel).

The relatively late fusion events (after 15 min p.i.) suggested that the penetration of SFTSV is likely to occur in LEs.[30] To confirm this possibility, we detected the pH threshold for SFTSV fusion using LysoSensor yellow/blue DND-160, which is a low-pH probe dye with fluorescence excited at two channels that may be used to measure the pH of acidic organelles (Figure 7C).[32] The QDs-DiD-SFTSV infected cells were labeled with LysoSensor yellow/blue, and the fluorescence signals of QDs (red), DiD (green) and the dye (rose and blue) were monitored simultaneously. The ratio R of rose to blue fluorescent intensity of LysoSensor yellow/blue, which colocalized with both the red (QDs) and green signals (DiD), was analyzed. Further, the pH at which fusion occurred was further calculated according to a pH standard curve (data not shown).[32] It was shown that the pH threshold of SFTSV penetration was mainly between the range of pH5.0–6.0, with an average fusion pH value of 5.6 (n = 68, Figure 7D), which is in line with the pH of LEs. Further colocalization analysis of five randomly selected cells showed extensive fusion of QDs-DiD-SFTSV with Rab7+ LEs; in contrast, fewer viruses fused with Rab5+ EEs (Figure 7E, white arrows). Taken together, these results suggest that the majority of SFTSV fusion events are triggered by the low pH (of ≈5.6) in LEs.

3. Discussion

Viral infection is a multi-step process involving extensive and elegant cellular interactions. Here, we adopted sensitive fluorescent labeling approaches and QD-based SPT for real-time visualization of the entire process of the SFTSV entry. These technologies allow us to monitor the infection dynamics and virus–host interactions, even the transient interactions and disassembly during entry, in living cells at single particle level in in situ.[22,33] In this study, Vero cells were used because they are permissive for SFTSV infection[34] and are suitable for living cell tracking due to their good adhesion to cell culture dishes. Vero cells were derived from African green monkey kidney,[35] and in vivo model showed that SFTSV replicated in kidney cells in mice and induced pathological lesions in kidney in macaques.[36,37] But the biological relevance of the current study to what happens in patients still awaits future confirmation.

Viruses have developed versatile strategies to overcome cellular barriers for the initiation of infection. Different viruses exploit various pathways for internalization into cells; in addition, the same virus may enter different cells using distinct routes.[17] Unlike the other two phleboviruses, UUKV and RVFV, which utilize clathrin-independent endocytosis to invade the host cells,[14,17] SFTSV was found, in the present study, to hijack the clathrin-mediated endocytosis machinery to invade host cells. This classical entry pathway is also utilized by other bunyaviruses, such as orthobunyaviruses, nairoviruses, and hantaviruses.[3] Although the entry of SFTSV is mainly mediated by clathrin, but there still remain low amounts of productive infections in the CPZ and sucrose treated cells (Figure S2B, Supporting Information). In addition, it was also observed that some internalizing viruses were not colocalized with clathrin (Figure 1F). This implies that SFTSV may take multiple

Figure 6. Endosomes carrying SFTSV moving along microtubules. A,B) Dynamics of virus-containing endosomes transport in the cytoplasm. A) Tracking QDs-SFTSV (red) containing Rab5+ EEs (green) (white arrows) movement in the cell periphery. The trajectory (middle) and MSD-Δt plots (right) of EEs containing SFTSV were analyzed. The green (I) and blue (II) lines represent the restricted motion and directed motion, respectively. B) Tracking QDs-SFTSV (red) containing Rab7+ LEs (green) (white arrows) moving in the cell interior region. The trajectory (middle) and MSD-Δt plots (right) of EEs containing SFTSV moving were analyzed. The orange, blue and pink lines represent the trajectories. C) A series of images of Rab5+ endosomes (red) carrying SFTSV particles (blue) moving along the microtubule (green) toward the cell interior (0–459.8s).

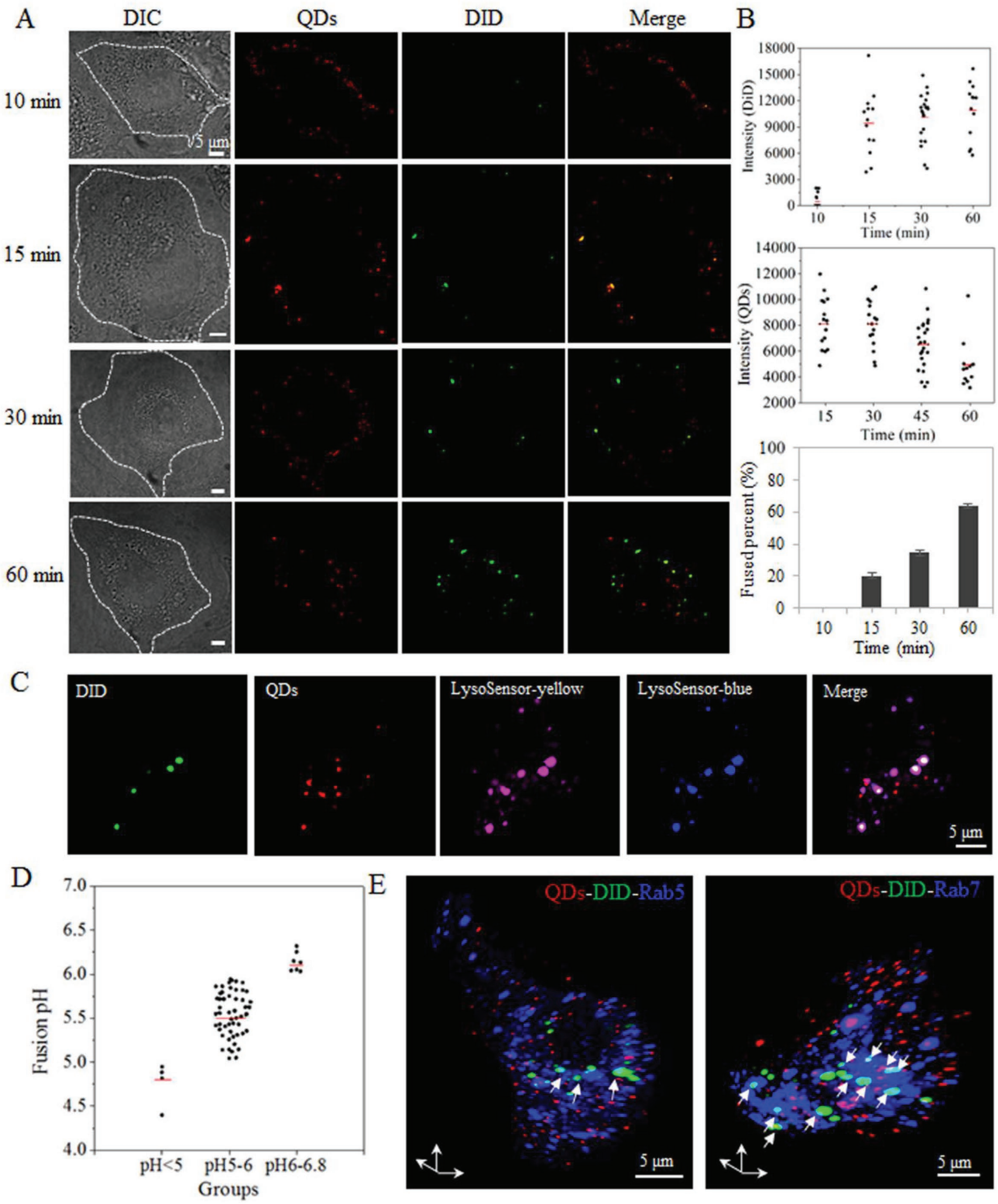


Figure 7. Acid-activated fusion of SFTSV in LEs. A) Characterization of QDs-DiD-SFTSV. Vero cells were infected with QDs (red) and DiD (green) dual-labeled SFTSV (QDs-DiD-SFTSVs) and images were acquired at 10, 15, 30, and 60 min p.i. The white dashed lines indicated plasma membranes. B) Time course of QDs-DiD-SFTSV fusion. Time-dependent change in fluorescence intensity of DiD (above), fluorescence intensity of QDs (middle), and the percentage of particles fused are shown; $n > 10$ cells in each group. C,D) The pH threshold for SFTSV membrane fusion. C) Confocal images of QDs-DiD-SFTSV in the cells labeled by LysoSensor yellow/blue were captured at 60 min p.i. The pH of the acid endosomes ($n = 68$) where SFTSV fusion was detected by LysoSensor yellow/blue. The bars show the means of the data (D). E) SFTSV fusion in the LEs. 3D confocal images of the QDs-DiD-SFTSVs with Rab5+ EEs at 45 min p.i. (left), with Rab7+ LEs at 60 min p.i. (right). DiD fluorescence spots rarely colocalized with Rab5, while most colocalized with Rab7 (white arrows).

pathways during infection, like the influenza virus, which not only uses clathrin-dependent, but also non-clathrin and non-cavolin-dependent endocytosis.[38] During clathrin-mediated endocytosis of SFTSV, clathrin is assembled on the plasma membrane for CCP formation; these CCPs then pinch off into the cytoplasm to form CCVs containing the internalized cargo (Figure 1C–G). In previous studies, with the aid of QD-based SPT, it was revealed that clathrin associated adaptors function to recognize cargo selectively and were recruited as the bridges for clathrin assembly to the virus binding sites, such as AP-2 for vesicular stomatitis virus (VSV) and epsin 1 for influenza A; then clathrin polymerization drives the progressive invagination of the pit, which is eventually released into the cytoplasm as an endocytic vesicle containing virions, through the co-action of the dynamin and actin.[25,39–41] Several viruses are able to induce the de novo assembly of clathrin at the site of virus binding, such as influenza virus, infectious hematopoietic necrosis virus and VSV.[25,41,42] Other viruses, such as dengue virus and canine parvovirus, seek pre-existing CCPs and then recruit more clathrin to form mature CCPs and further CCVs.[43,44] We found that SFTSV infection led to the recruitment of clathrin onto the plasma membrane and promoted the formation of CCVs (Figure 1C–E). Though a proportion of the membrane-bound SFTSV particles were found to be colocalized with clathrin coats on the cell surface, and then invaginated into host cells through detachment of CCPs from the plasma membrane (Figure 1F), but since virions were bound to the cell surface for 30 min at 4 °C during the QDs labeling process before observation, whether SFTSV particles bound to pre-formed CCP or they induced de novo CCP formation upon binding remains further investigation.

Similar to other bunyaviruses, such as UUKV, CCHFV, and LACV,[3] following their internalization via CCVs, SFTSV were found to be entrapped in larger intracellular vesicles. The typical morphologies of EE, MVE, and LE encompassing virions suggested that the internalized SFTSV were channeled through the endocytic pathway (Figure 2). Continuous tracing of labeled virions and endosomes suggested that SFTSV traffic through CCVs to Rab5+ EEs, and then to Rab7+ LEs during their journey in the endocytic machinery (Figure 3). Through real-time multicolor-tracking of viral particles, CCV and endosomes, we visualized the delicate processes of shedding off clathrin coats from the primary endocytic vesicles and delivery of virions to EEs (Figure 3A). Further, Rab5 was recruited from the cytosol to the cytosolic surface of the EE membrane, representing the maturation of EE from CCV.[29] The conversion between Rab5 and Rab7 is an essential step in LE formation: during this process, Rab7 is recruited to the EE by Rab5, and the activated Rab7 suppresses Rab5.[45] We also observed that viruses located in a hybrid Rab5/Rab7 endosome showed gradual removal of Rab5 and its replacement with Rab7 (Figure 3B). Taken together, our data strongly support a dynamic transport route from the CCVs to EEs, then to LEs for SFTSV cytoplasmic trafficking.

In order to reach the site of replication, numerous viruses manipulate the host's cellular cytoskeleton system, including actin filaments and microtubules, for efficient intracellular transportation.[41] Although the involvement of the cytoskeleton in certain bunyavirus infections has been suggested by using inhibitors, detailed investigation has not been performed.[20,21] Combining SPT with biochemical analysis, we were able to show that SFTSV initially experienced actin filament-dependent trafficking when passing through the plasma membrane and at the cell periphery region; this was followed by microtubule-dependent transport toward the interior of the cells. This “microfilament-microtubule relay” mechanism used by SFTSV is similar to that used by the influenza virus.[46] The actin filaments provide an essential driving force for viral entry,[47] For instance, during the entry of HIV-1, polymerization of actin is required to assemble co-receptors (CD4 and CXCR4) at the plasma membrane, which promotes virus binding; meanwhile the cortical actin poses a barrier for early viral postentry events until depolymerization of actin through activated actin-interacting protein cofilin.[48] During transport process of influenza virus, myosin VI, an actin-based motor, is colocalized with CCP/CCV and mediates CCV movement along the actin filaments away from the periphery region.[46] After CD and BDM treatment, SFTSV infection was significantly reduced, and particles were mostly observed to be continuously “trembling” in a limited area near the cell membrane (Figure 5 and data not shown). These findings indicate that the actin filaments and their motor myosin play a crucial role in the clathrin-mediated internalization of SFTSV at the cell periphery. Real-time tracking also showed that SFTSV particles were transported along the microtubules in the cell interior (Figure 4D); however, after incubation with nocodazole, SFTSV particles underwent short-distance directed movement without penetrating deeper into the cell interior (Figure 5). These observations indicate that the virus switched from trafficking along the actin filaments at the cell periphery to microtubule-based movement in the cell interior region. It was of interest to further explore the mechanism of the switchover of SFTSV transport from actin filaments to microtubules. By using multi-color tracing, we directly visualized SFSTV-carrying endosomes moving along the microtubules toward the cell interior (Figure 6). Endosomes may be transported actively along microtubules; this is dependent on dynein (plus-end-directed) and kinesin (minus-end-directed) motors, which provide two opposing forces that drive endosomes moving bidirectionally.[49] Here, we confirmed, via dynamic analysis, that the virus-containing endosomes experienced bidirectional directed cytoskeleton-dependent motion, suggesting that both dynein and kinesin were engaged in the transport of SFTSV-containing endocytic vesicles.

As the ultimate and an essential step of the entry process, endocytosed viruses penetrate the membrane of endosomes to release their genome and accessory proteins into the cytosol. Membrane fusion with EEs or LEs is triggered by the acidic environment, which activates the virus envelope fusion proteins. For viruses that require a relatively high pH threshold for fusion, such as VSV, membrane fusion is likely to occur in EEs.[44] For viruses with a lower pH requirement for fusion, such as influenza virus and some bunyaviruses, this event occurs in LEs.[30] The fusogenic abilities and dynamics of numerous viruses have been investigated by various in vitro approaches; these include measurement of the effect on infection of the addition of NH_4Cl at different times after endocytosis, or observation of the formation of syncytia in cell with excessive virus binding onto the cell surface. Viruses fusing in EEs, such as VSV and the Semliki forest virus, require a pH threshold of 6.3–6.1 and become NH_4Cl-insensitive within 5–8 min postentry, suggesting that the viruses have already passed through the acid-activated step.[44,50] In contrast, for viruses penetrating within LEs, such as UUKV, fusion is triggered at a pH below 5.8, with an optimum at 5.4, and occurs at 20–40 min postinternalization.[14] For SFTSV, it was reported that syncytia in SFTSV-infected cells formed at 1 h after treatment with buffers below pH 6.0.[34] Using QD and DiD dual-labeled virions, we provided direct evidence that membrane fusion of SFTSV started at 15 min, and reached a half level within 60 min (Figure 7). Similar observations have been reported for Ebola virus and influenza virus by direct visualization.[26,51] The colocalization analysis of QDs-DiD-SFTSV with endosomes also confirmed that extensive fusion of QDs-DiD-SFTSV occurred in Rab7+ LEs (Figure 7E). Direct measurement of the fusion pH of SFTSV with endosomes in living cells by a low-pH probe showed that the mean fusion pH value is ≈5.6 (Figure 7D). These data provide direct evidence for low-pH-dependent SFTSV penetration from LEs.

4. Conclusion

In conclusion, in the present work, the entire entry processes of SFTSV have been depicted systematically via QD-based SPT combined with biochemical analysis and ultrastructural observation (**Figure 8**). We have shown that, unlike UUKV and RVFV, the SFTSV entry process is clathrin-dependent. SFTSV is internalized into the host cell via the recruitment of clathrin onto the plasma membrane for CCP formation. Then the virus-containing CCVs deliver SFTSV particles and membranes to Rab5+ EEs, and further to Rab7+ LEs, a common mechanism of cytoplasmic delivery also shared by other bunyaviruses, such as UUKV, RVFV and CCHFV.[14,15,19] The intracellular trafficking of SFTSV is dependent on the two different kinds of cytoskeleton in sequence: first on actin filaments at the cell periphery, subsequently switching to microtubules toward the cell interior. It has been demonstrated that cytoskeletons are also involved in hantavirus and CCHFV entry, suggesting that cytoskeleton might be generally involved in bunyavirus transport.[20,21] For the vast majority of virions, acid-activated penetration was triggered at pH5.6 in LEs. The results enrich our understanding of the entry mechanisms of bunyaviruses, and provide potential targets for SFTS prevention and control.

5. Experimental Section

Cells and Viruses: Vero (ATCC CCL-81) used in this study were cultured in Dulbecco's Modified Eagle's Medium (DMEM, Gibco) supplemented with 10% fetal bovine serum at 37 °C to achieve 75% confluence. SFTSV HBMC5 was propagated in Vero cells at 37 °C, and virus stocks were maintained at −80 °C in the laboratory.[52]

Purification of SFTSV: SFTSVs were propagated in Vero cells and harvested at 7 days p.i. The cell debris was removed by centrifugation at 8000 × *g* at 4 °C for 30 min, and then by passage through a 0.45 µm filter (Millipore). The supernatant was used for virus biotinylation, as follows: roughly 300 mL of virus supernatant (≈10^7 $TCID_{50}$ mL^{-1}) was reacted with 300 µL of sulfo-NHS-LC-biotin (0.01 mg $µL^{-1}$, Thermo) at room temperature for 2 h. The supernatant of biotinylated SFTSV and unbiotinylated SFTSV was then purified by tangential-flow ultrafiltration. Briefly, the prepared supernatant (300 mL) was added into the reservoir and pumped through an ultrafiltration membrane cassette with a molecular weight cut-off of 100 kDa (Sartorius, Vivaflow 50R). The recirculation rate was maintained at 300 mL min^{-1} by a peristaltic pump. This process was continued until 30 mL of retentate finally remained in the reservoir. The ultrafiltered virus supernatant was negatively stained and examined by TEM. During this process, unbound biotin and impurities were removed, and the purified bio-SFTSV and SFTSV particles were obtained and stored at −20 °C until use.

Labeling of SFTSV: For QDs labeling, SFTSV was labeled through the biotin–streptavidin interaction to link SA-QDs to biotinylated SFTSV. The method consisted of standard procedures: 100 µL bio-SFTSV was bound to a monolayer of Vero cells for 25 min at 4 °C and then washed three times with pre-chilled PBS containing 0.1% bovine serum albumin (BSA). The plates were then incubated with 100 µL of 5×10^{-9} M SA-QDs 605/705 (Wuhan Jiayuan) at 4 °C for 15 min. After three additional washes with PBS (with 0.1% BSA), the QDs labeled SFTSV (QDs-SFTSV) particles were obtained.

For dual-labeling viral envelope with DiD and QDs (QDs-DiD-SFTSV), purified SFTSV was incubated with biotin for 30 min, followed by the addition of DiD (50×10^{-6} M) at room temperature for 1.5 h. The unbound biotin and DiD were removed by a NAP-5 desalting column (GE Healthcare), as described in the product manual. The DiD-labeled bio-SFTSV was then labeled with SA-QDs as described above.

Immunofluorescence Assay (IFA) and Virus Titration: For analysis of labeling efficiency, the purified bio-SFTSV or SFTSV was bound to a monolayer of Vero cells, respectively. The viruses were labeled by SA-QDs as described above; then, the QDs-SFTSV was immunostained by IFA. The infected cells were fixed and incubated with anti-NP antibody prepared in the laboratory,[52] then stained with AF 488-conjugated goat anti-rabbit IgG.[45] The cells were imaged by spinning-disk confocal laser scanning microscopy (CLSM) with two kinds of fluorescence signals, and the colocalized efficiency of the two kinds of signals was calculated as labeling efficiency.

For the infectivity analysis of labeled virus, Vero cells were incubated with 100 µL of purified SFTSV, bio-SFTSV or QDs-SFTSV (100 µL bio-SFTSV labeled with QDs), and the infection supernatant was harvested at 48 h p.i Viral titers were determined by end-point dilution assays (EPDAs).[52]

Plasmid Transfection: Vero cells were transfected with plasmids pEGFP-Cla, pEGFP-Cav, pEGFP-Rab5/pdsRed-Rab5, pEGFP-Rab7, pEGFP-LifeAct, and pEGFP-MAP4 using Lipofectamine 3000, according to the manufacturer's protocols, for labeling clathrin, cavolin, EEs, LEs, microfilaments and microtubules, respectively.

TEM: For examination of the morphology of the purified SFTSV and bio-SFTSV, the purified viral particles were negatively stained with 2% (w/v) phosphotungstic acid (PTA, pH 6.8) and observed using a 200 kV transmission electron microscope (FEI Tecnai G^2 20 TWIN). For thin-section EM, after exposure to virus (at an MOI of 20), cells were fixed and treated as previously[53] and examined by TEM.

Inhibition of SFTSV Infection and Western Blotting Analysis: Several inhibitors (Sigma) of various cellular entry pathways were used to investigate the entry mechanisms of SFTSV. Here, CPZ (0–35 × 10^{-6} M) and sucrose (0–100 × 10^{-6} M) for inhibiting clathrin-mediated endocytosis; nystatin (0–50 × 10^{-6} M) and PMA (0–15 × 10^{-6} M) for blocking caveola-mediated endocytosis; RAC-1 (0–200 × 10^{-6} M) and IPA3 (0–30 × 10^{-6} M) for inhibiting micropinocytosis; nocodazole (0–10 × 10^{-6} M) for disrupting the microtubules; CD (0–20 × 10^{-6} M) for blocking microfilament-dependent transport; and CQ (0–60 × 10^{-6} M) and NH_4Cl (0–80 × 10^{-6} M) for disrupting the acidification, were applied. Cells were treated with DMSO/PBS (control, 0 × 10^{-6} M) or with each inhibitor for 2 h prior to SFTSV infection. Treated cells were infected with SFTSV (5 MOI) for 2 h at 37 °C in the continued presence of the inhibitors. Then, cells were washed three times with PBS, then incubated at 37 °C, and harvested at 48 h p.i. for viral titration by IFA or for Western blotting analysis. The cytotoxic effects of the inhibitors were determined using the cell counting kit-8 (MCE). The results showed that the doses of the inhibitors used in this study did not cause cell toxicity (data not shown).

Viral protein expression levels in infected cells were analyzed by Western blotting. The anti-NP antibody served as the primary antibody, and Pierce horseradish peroxidase (HRP)-conjugated polyclonal goat anti-rabbit IgG was used as the secondary antibody (Thermo). Internal control reactions to detect *β*-actin were carried out simultaneously.

Live Cell Fluorescence Imaging: For single-color tracking, Vero cells infected with QDs-SFTSV were observed by CLSM immediately after infection. For multicolor observation, after specific cellular components were labeled by corresponding plasmid transfection for 36 h, the cells

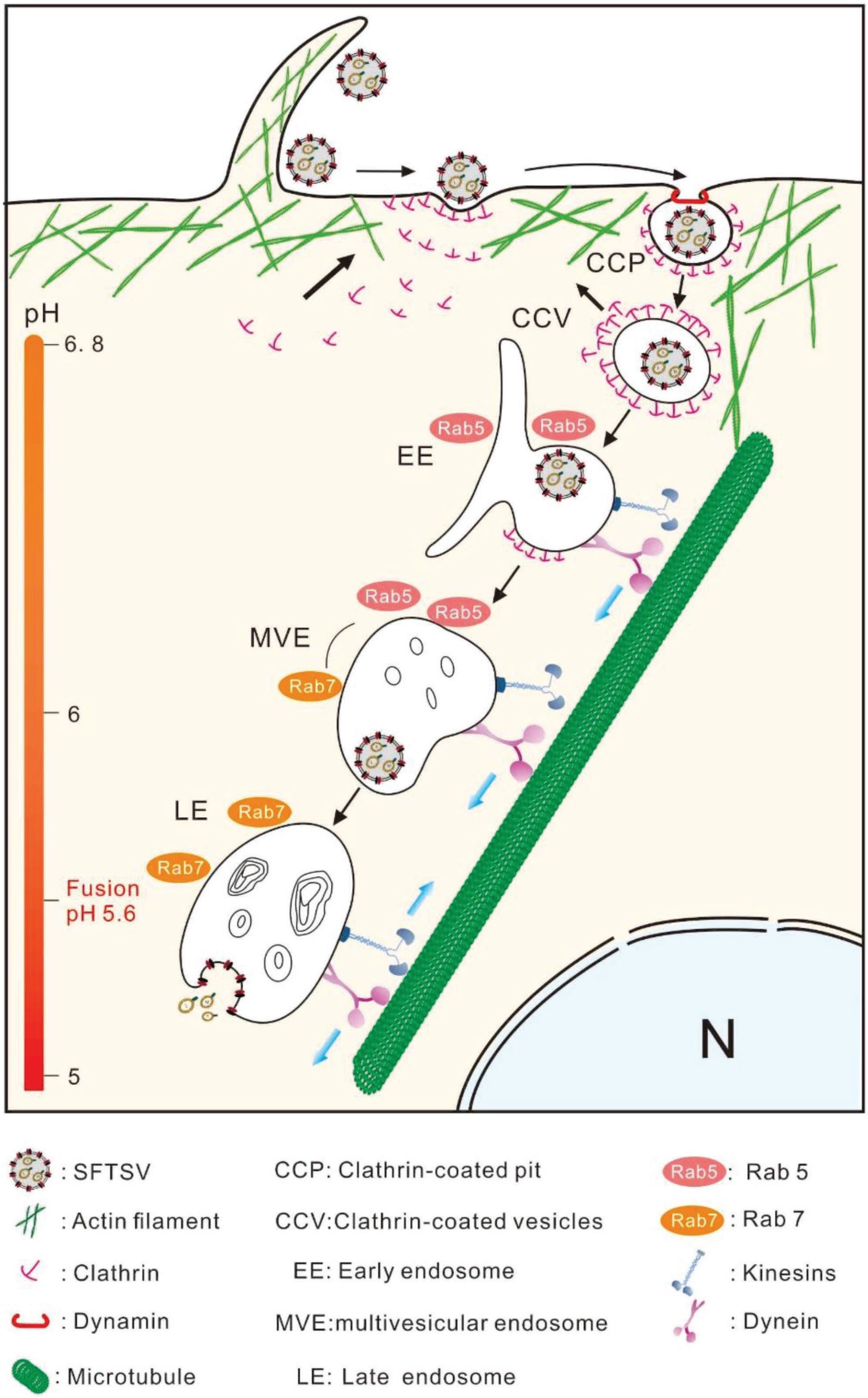


Figure 8. Entry model of SFTSV into host cells. Internalization of SFTSV into host cells was initiated by recruiting clathrin onto cell membrane for clathrin-coated pits formation and further pinching off from plasma membrane to form discrete vesicles. These clathrin-coated vesicle carriers further delivered virions to Rab5+ early endosomes, and then to Rab7+ late endosomes. Then virion-carrying endocytic vesicles was transported on actin filaments at cell periphery, and then on microtubules toward cell interior. The final fusion events occurred at ≈15–60 min postentry, and were triggered by acidic environment at ≈pH5.6 within the late endosomes.

were infected with QDs-SFTSV. All observations were performed at 37 °C. Fluorescence images were obtained using a spinning-disk confocal microscope (PerkinElmer UltraVIEW VoX) equipped with a cell culture system (INUBG2-PI). LysoSensor yellow/blue, GFP/AF 488, dsRed/QDs 605 and QDs 705/DiD were excited with lasers of wavelengths 405, 488, 561, and 640 nm, respectively. The fluorescence signals were separated using band-pass emission filters of 447/60 and 525/50, 525/50, 605/20, and 685/40 nm, respectively.

Measurement of SFTSV Fusion pH: Vero cells were infected with QDs-DiD-SFTSV at 37 °C for 55 min, then incubated with 10×10^{-6} M LysoSensor yellow/blue DND-160 (Invitrogen) for labeling acidic organelles at 37 °C for 5 min. The cells were washed twice and immediately imaged by CLSM. The fluorescence signals of QDs, DiD, and LysoSensor were observed. LysoSensor data were recorded at both excitation/emission wavelengths of 360/440 and 360/540 nm. The co-localization of QDs, DID, and blue/rose endosomes represented the fusion of the virus occurred within these endosomes; then the fluorescent intensity ratio of these LysoSensor signals was calculated as $R = (F_1–B_1)/(F_2–B_2)$, where F_1 and F_2 are the fluorescence intensities at 360/540 and 360/440 nm, respectively, and B_1 and B_2 are the corresponding background values determined on the same images. The endosomal pH was calculated through the standard curve between R and pH established as previously described.[32] In brief, Vero cells were incubated with 10×10^{-6} M LSYB for 5 min and then permeabilized with 10×10^{-6} M monesin dilution in MES buffer (115×10^{-3} M KCl, 5×10^{-3} M NaCl, 25×10^{-3} M MES, and 1.2×10^{-6} M $MgSO_4$) with various pH from 4.0–7.0 for 10 min. The data of R and pH was fit to a Boltzmann Sigmoid function.

Image Analysis: Images were denoised by processing with a Gaussian filter. To quantify the colocalization extent of the two fluorescent signals, for example QDs-SFTSV (red) and Rab5/Rab7 (green), intensity correlation analysis (ICA) was performed by Volocity 2.6 (Colocation Threshold plugin for 3D image) and image J (Colocation Threshold plugin for 2D image).[28,54] To analyze the location of the clathrin in the cells, the cell boundaries were first recognized according to the merged images of the bright field images and the fluorescent images, and then the proportion of the fluorescence signal dots (measured by Analyze particle plugin in image J) on the boundary lines (clathrin on the membrane) to the total signal dots in the whole cells was calculated. To construct the 3D confocal images, the three-channel Z-stacks were recorded with the gap of 0.3 µm which were processed through 3D visualization-assisted analysis (Vaa3D). The movement of labeled particles was tracked and analyzed using Image-Pro Plus 6.0. The trajectories of QDs-SFTSV, which consisted of the traveling distances between consecutive frames, were generated. The time trajectories of the velocity were indicative of the instantaneous velocity of a viral particle in living cell. MSD was calculated by an equation using a compiled program based on MATLAB.[55] The motion mode was determined by fitting MSD and Δt to the following functions: $\mathrm{MSD} = (V\Delta t)^2 + 4D\Delta t +$ constant (directed diffusion), $\mathrm{MSD} = 4D\Delta t^{\alpha} +$ constant (restricted diffusion), and $\mathrm{MSD} = 4D\Delta t +$ constant (free diffusion), where D represents the diffusion coefficient, V represents the mean velocity and α is a constant less than 1. For the intensity analysis of the fluorescence (QDs and DID), the image signals were normalized against background signals ($F–B$, F: the fluorescence intensities of the target signals, B: the corresponding background fluorescence intensities).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

clathrin, membrane fusion, quantum dots, severe fever with thrombocytopenia syndrome virus (SFTSV), single-particle tracking

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